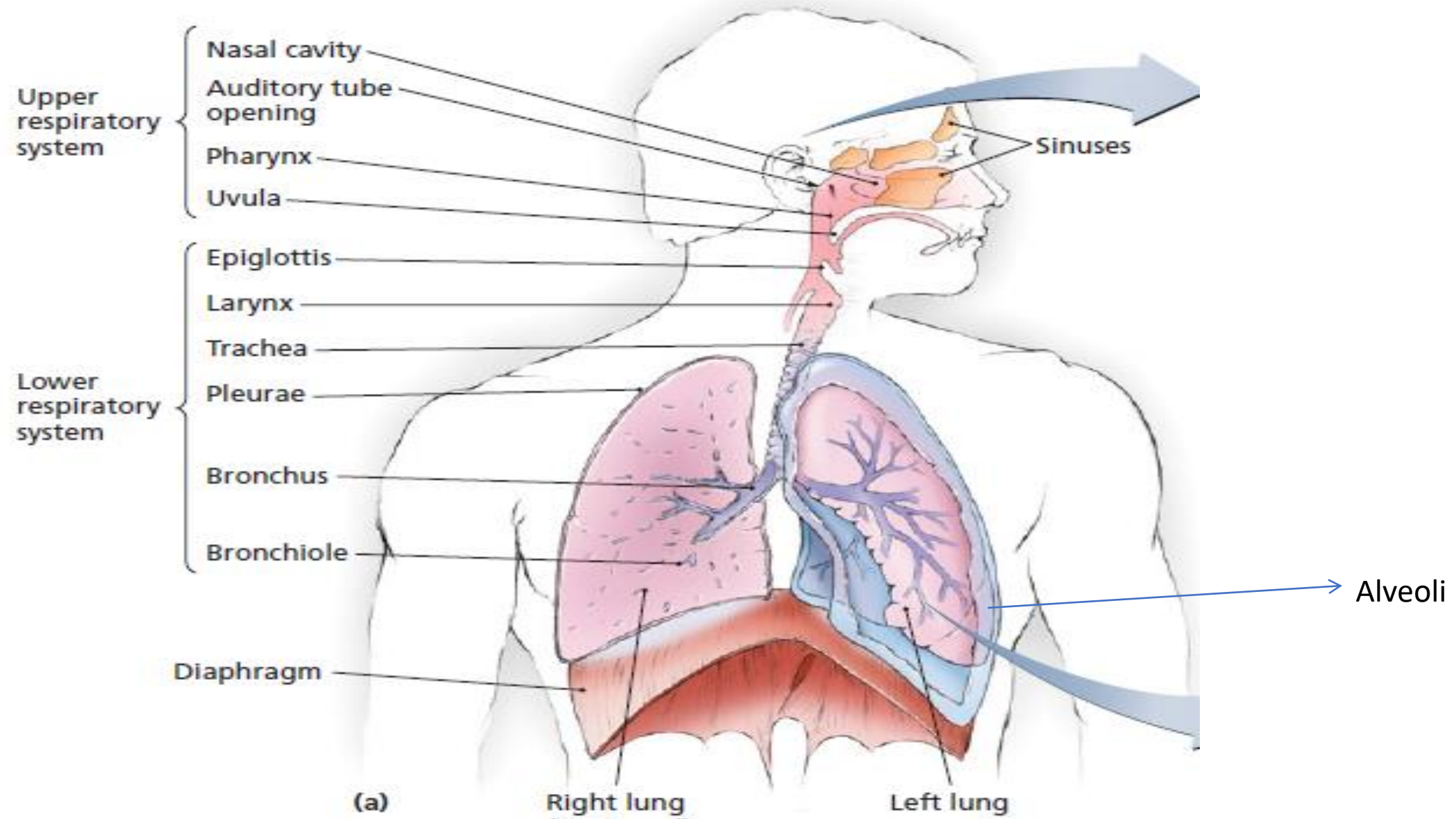


RESPIRATORY TRACT INFECTIONS (RTI)

Structures of Respiratory system



Microbial Diseases of the Respiratory System

1. Normal Microbiota of the Respiratory System

The lower respiratory system normally lacks microorganisms

This is because of the presence of :-

- the ciliary escalator,
- secretory antibodies,
- phagocytic

NORMAL FLORA OF THE RESPIRATORY TRACT

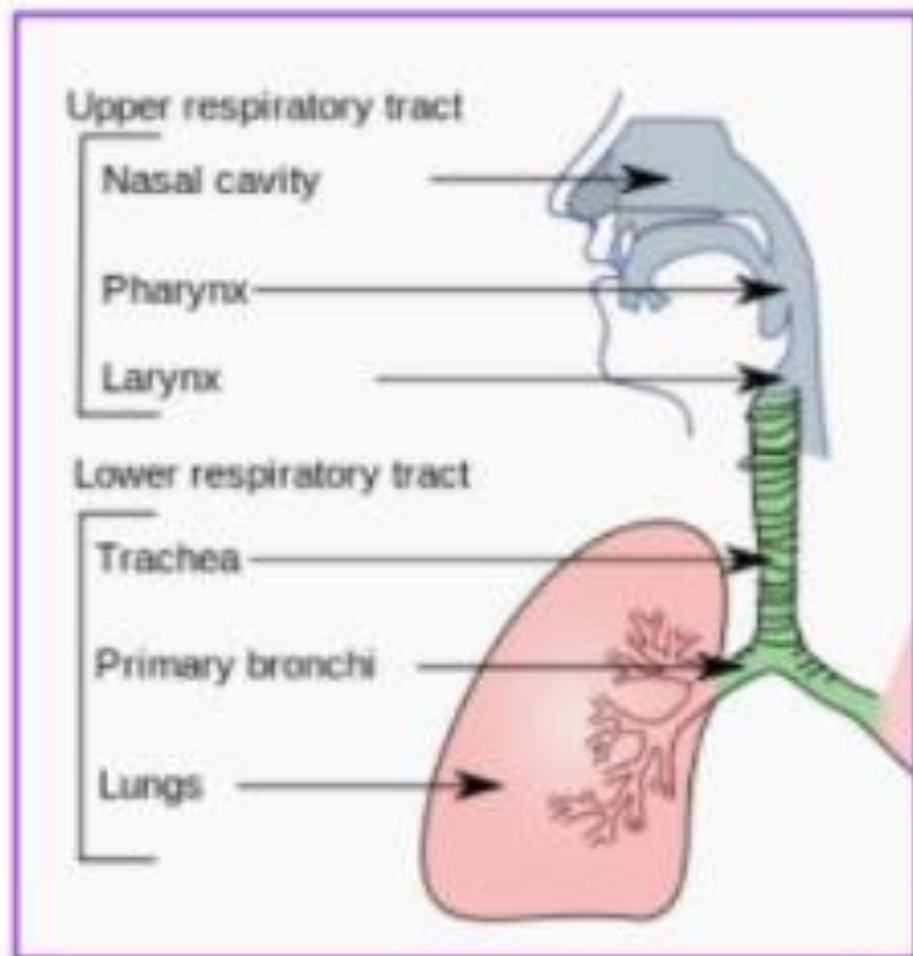


Lower respiratory tract:

Sinuses, trachea, bronchi, bronchioles, and alveoli are sterile.

Upper respiratory tract:

Nasal cavity, nasopharynx and pharynx (throat) are colonized by many bacterial species.



NORMAL FLORA OF THE RESPIRATORY TRACT

Upper respiratory tract:

- The moist warm mucous membrane in this area provide a very good environment for the growth of many varied population of microorganisms.
- These organisms include: *Diphtheroids*, *Staphylococcus spp.*, *Micrococcus spp.*, and *Streptococcus spp.*

Some people are carriers for some potentially pathogenic organisms such as:

Nasopharynx: *S.pneumoniae*, *S.pyogenes*.

Nose: about 20% of people carry *S.aureus* in their nose.

The normal microbiota play an important role controlling infections by:-

1. removing nutrients (compete for nutrients)
 2. and releasing substances that inhibit the growth of pathogens
- Some of them are opportunistic pathogens.

Respiratory tract infections (RTI)

- Upper respiratory tract infection (URTI)
 - Viral, bacterial
- Lower respiratory tract infection (LRTI)
 - Bacterial, viral, fungal, parasite

Bacterial infections of Upper Respiratory Tract

Upper respiratory tract infections

- Most common infections both in children/adults
 - Most caused by viruses & usually nothing more than irritation
- Direct invasion of respiratory epithelium results in symptoms corresponding to the area(s) involved
- Include tonsillitis, pharyngitis, laryngitis, sinusitis, influenza, otitis media, common cold
- Symptoms: cough, sore throat, runny nose, headache, nasal congestion, low grade fever, facial pressure, sneezing
- May spread down and cause more severe infections

DIPHTHERIA

- Caused by *Corynebacterium diphtheriae*

- **Characteristics** —

Club-shaped gram-positive rods arranged

in V or L shape, Aerobic, non-spore-forming organism

Habitat and Transmission—Habitat is the human throat.

Transmission is via respiratory droplets.

Virulent factors:- Organism secretes an exotoxin that inhibits protein synthesis

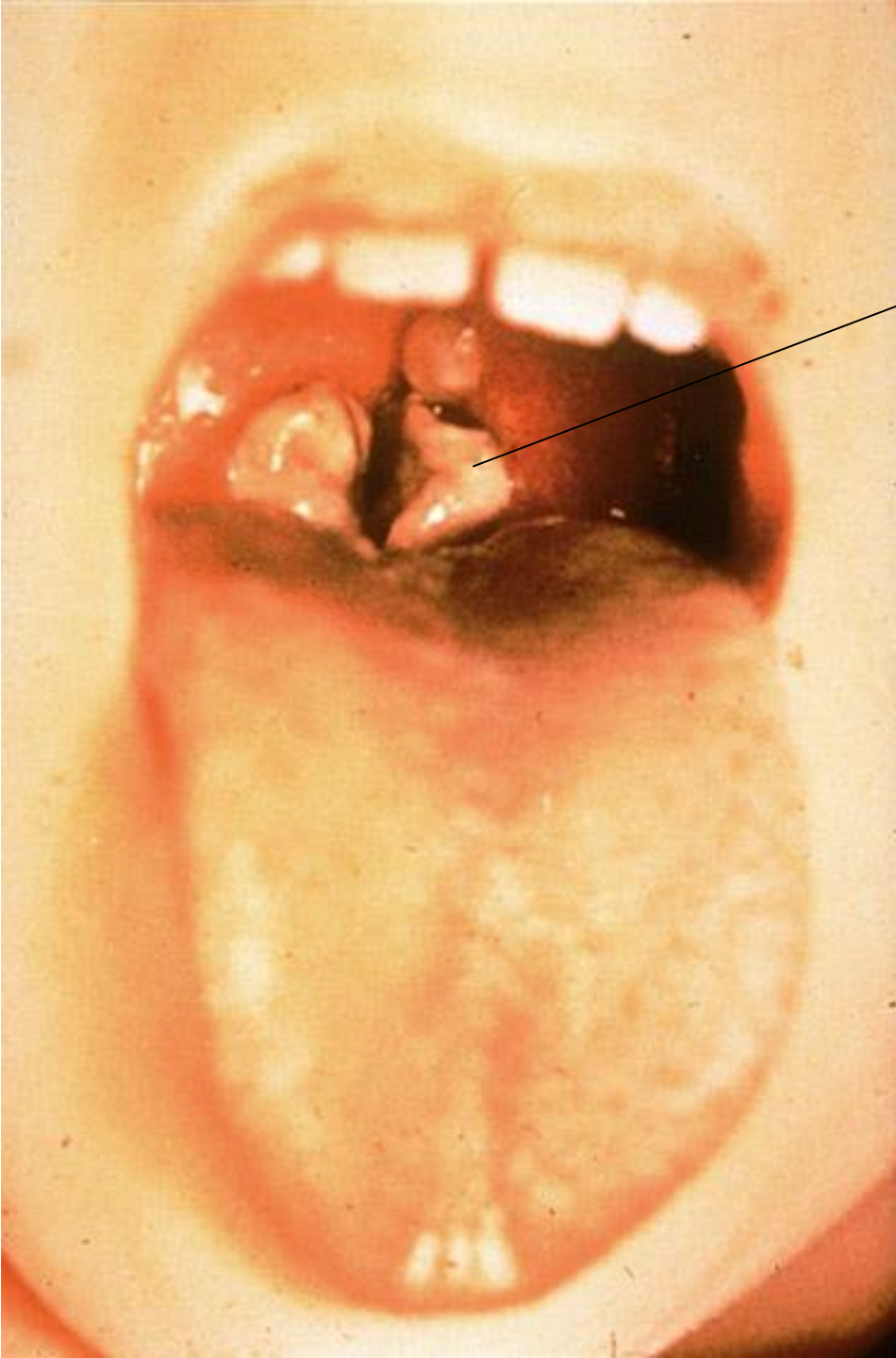


pathogenesis

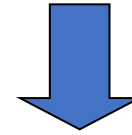
► Organism multiply on epithelial cells of pharynx, tonsils or larynx
► Release toxin and cause local damage, malaise, sore throat, exudative pharyngitis, and form **Pseudomembrane** (exudative structure) will be formed on pharynx which contain bacteria, lymphocytes, plasma cells, fibrins, and dead cells

Pseudomembrane can cover tonsils, Uvula, palate or extend up to nasopharynx or down to pharynx then the patient may develop breathing obstruction

- Toxemia can cause damage of heart, coma and death
- Don't attempt to remove the membrane it will result
→ **bleeding**



**Pseudomembrane
on tonsils**



**can lead to
respiratory
blockage.**

Laboratory diagnosis

✦ Based on clinical grounds:- patient should be treated immediately because of the effect of powerful exotoxin and the patient may die until the laboratory result comes

- Specimens:- throat/nasopharyngeal swab, skin swab
- Microscopy:- Gram positive rods and polymorphic, form angles

Albert stain:- stain green, appear beaded (energy storing granules)

- Culture on to different selective media like

Cysteine tellurite agar

- **Toxigenicity test**

Testing for exotoxin production (Elek test)

- **Treatment**

Focused on:

Rapid neutralization of toxins then killing of toxin producing bacteria

- Antitoxin to neutralize exotoxin
- Penicillin, or erythromycin to eliminate bacteria.

Prevention

Administration of vaccine and booster shots (at ages of 2,4,6,15 to 18 months, 4 to 6 years and then every 10y)

Streptococcus pyogenes

Disease:-

- Suppurative (pus-producing) diseases (e.g., [pharyngitis](#) and cellulitis);
- nonsuppurative (immunologic) diseases (e.g., rheumatic fever and acute glomerulonephritis).

Characteristics—

- Gram-positive cocci in chains.
- β -Hemolytic colonies.
- Catalase-negative.
- Bacitracin-sensitive



Virulence Factors

- have a number of structures, enzymes, and toxins:- namely
 1. **M proteins** destabilize complement component **C3b**, thereby interfering with opsonization and lysis.
 2. The **hyaluronic acid capsule** may “camouflage” the bacterium from phagocytes.
 3. **Streptokinases** are enzymes that break down blood clots, presumably enabling group A streptococci to spread rapidly through damaged tissues.
 4. **C5a peptidase** is an enzyme that breaks down complement protein C5a, which is a chemotactic factor. With this enzyme, *S. pyogenes* decreases the movement of leukocytes into the site of infection.
 5. **Pyrogenic (erythrogenic) toxins** stimulate leukocytes to release cytokines that in turn stimulate fever, rash, and shock.
 6. **Streptolysins** lyse erythrocytes, leukocytes, and platelets

Strep throat," or streptococcal pharyngitis/ tonsillitis

S/S:- The back of the pharynx appears red, with swollen lymph nodes and purulent (pus-containing) abscesses covering the tonsils



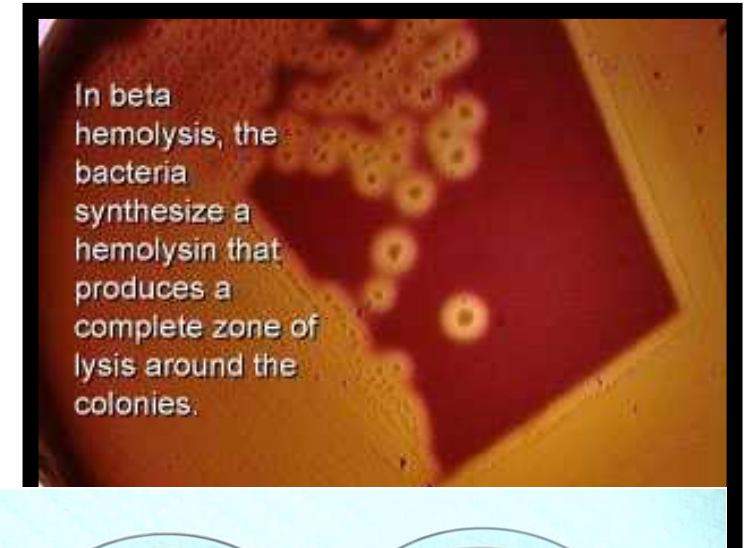
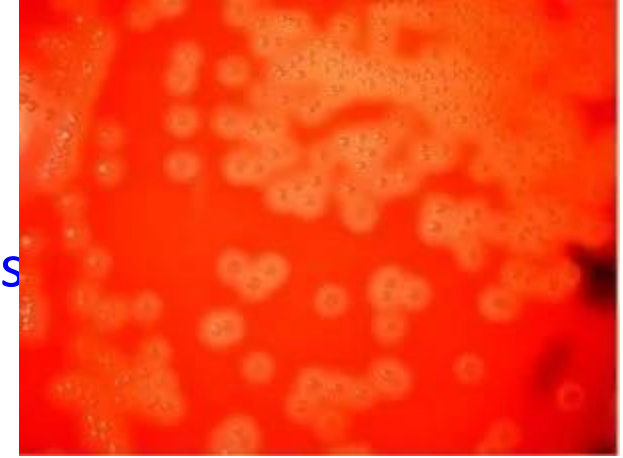
Pharyngitis caused by *Streptococcus pyogenes*. Note inflamed pharynx, tonsils, and palatal petechiae

Long-Term Complications of Group A Infections

- **Rheumatic fever** – follows overt or subclinical pharyngitis in children;
carditis with extensive valve damage possible, arthritis, fever
- **Acute glomerulonephritis** – nephritis, increased blood pressure,
occasionally heart failure; can become chronic leading to kidney failure

Laboratory diagnosis

- **Microscopy:** gram stain (gram (+) cocci)
- **Culture:** enriched media (blood plate, β – hemolysis)
- **Bacitracin** sensitive
- **Antigen detection**
 - Immunological assays for group specific CHO in Bacterial cell wall (Ag: CHO (+))
- **Antibody detection:** anti-SLO (ASO) test
 - Presence of Ab's to streptolysin-O confirms
 - Rheumatic fever
 - Acute Glomerulonephritis
- **Treatment**
 - **Penicillin**, erythromycin, cephalosporin



Prevention

- Early detection and treatment
- Patient with history of RF should take prophylaxis treatment before doing certain procedures (dental extraction) which induce bacteremia → endocarditis

SINUSITIS

Sinusitis is inflammation of the paranasal sinuses

Clinical Manifestations

- purulent nasal discharge, nasal congestion, facial or sinus pain, decreased sense of smell, and fever.
- Headache and malodorous breath may be present.

Pathogens

- *S. pneumoniae*, *H. influenzae*, *M. catarrhalis*, *Staphylococcus aureus*
- In immunocompromised patients and diabetics, sinusitis caused by fungi such as *Aspergillus* or *Mucor* may occur.

Treatment

- **Amoxicillin** is the drug of choice, but if resistance is a concern, then **amoxicillin-clavulanate (Augmentin)** is used

OTITIS MEDIA

- is an infection of the middle ear caused by either viruses or bacteria. Otitis media can be either acute or chronic

Clinical presentations

- ear pain and pressure. In infants, the ear pain may manifest as ear pulling. Patients may also complain of decreased hearing and fever

Pathogens

- Streptococcus pneumoniae is the most common cause.
- Nontypeable strains of Haemophilus influenzae and Moraxella catarrhalis are also common causes

Diagnosis:- culture of ear discharge

Treatment:- Amoxicillin orally is usually the drug of choice

. In cases of bacterial resistance, amoxicillin-clavulanate (Augmentin) may be used

Viral agents

1. Corona viruses

- *Corona* = crown like structure, pleomorphic, spiked
- Non-segmented, linear, ssRNA
- Usually confined to URT, causes common cold
- Highly species specific, common in children
- Patchy destruction of cilia and ciliated epithelial cells → development of disease
- High frequency of recombination during replication - evolution of new strains

Rhinoviruses (common cold virus)

- It is the most important causative agents of common cold and upper respiratory infections
- The virus infect the upper respiratory tract, including some or all of the following structures:
nose, throat, sinuses, eustachian tubes, trachea, and larynx
- There are more than 100 serotypes (no vaccine)
- Grow best at 33 degree Centigrade (has predilection to nasal mucosa, cooler environment)
- The virus is transmitted through direct contact via infected hands, and fomites, inhalation of infectious droplets

Clinical Manifestations

Clinical manifestations include nasal congestion, decreased sense of smell, rhinorrhea (watery nasal discharge without purulence), and sneezing. Patients also complain of general malaise and sore throat. In some cases, headache may also be reported

.

- other viruses associated with the common cold are coronaviruses, influenza viruses, respiratory syncytial virus, adenoviruses, enteroviruses and parainfluenza viruses
- Infected cells release bradykinin and histamine, which causes a runny nose
- Multiple serotypes exclude prior immunity

Treatment and prevention

- No specific antiviral therapy is available.
- Inhaling hot, humidified air/steam may help
- Paper tissues impregnated with citric acid may also limit rhinovirus spread
- Nasal vasoconstrictors may provide relief, but their use may be followed by rebound congestion and a worsening of symptoms
- Washing hands and disinfecting contaminated objects help prevent spread
- Vaccines appear impractical because of the large number of serotypes

Parainfluenza virus

- Respiratory virus
- Cause:- Croup
- **Croup** is an inflammation of the larynx, trachea, and large bronchi (laryngotracheobronchitis)
- There are **four serotypes** associated with human disease
- **1,2,3**, causes severe lower respiratory tract infections and associated with **laryngotracheobronchitis (croup)**
- Type 4 causes only mild upper RTI

- PIV Infect epithelial cells of URT and cause cell lysis

Clinical Syndromes

- In **infants** it may cause **severe infection** than in adults like bronchiolitis, pneumonia and **croup (laryngotracheo bronchitis)**
- **Croup (cough that sounds a bark of dog or seal)**
- results from subglottal swelling (narrowing of airway), which may close the air way

- When a **cough** forces air through this narrowed passage, the swollen vocal cords produce a noise similar to a **seal barking**
- The principal differential diagnosis is epiglottitis caused by *H. influenzae*

EPIDEMIOLOGY

- Host range is limited to humans.
- Reinfection can occur later in life.

Transmission

Inhalation of large-droplet aerosols.

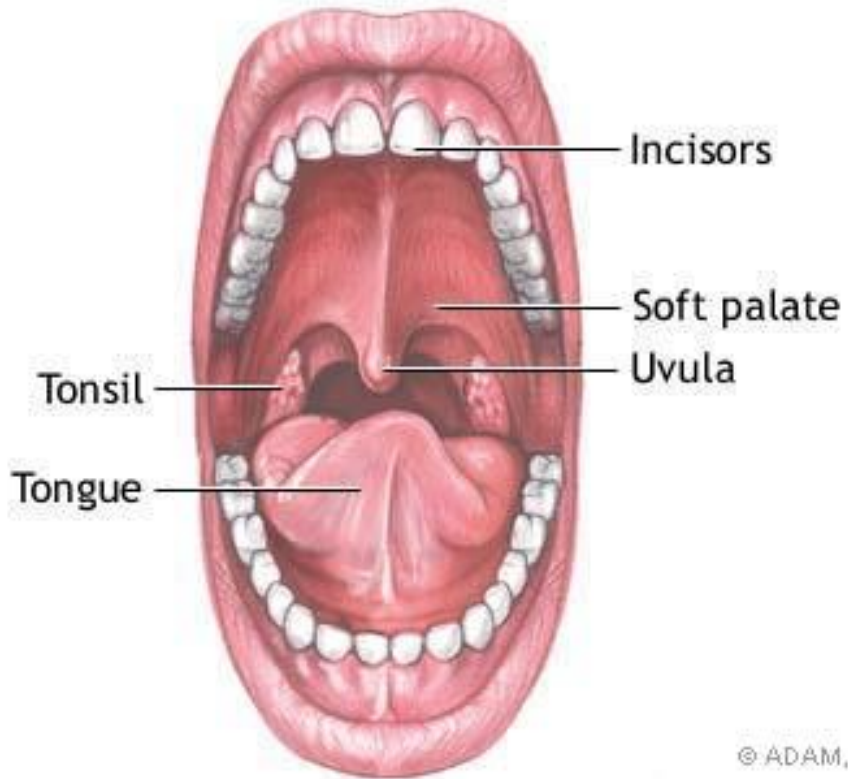
Treatment prevention and control

- No specific antiviral agent
- Administration hot steam and Monitoring of air way.
- Intubation (placement of a flexible plastic tube into the trachea windpipe to maintain an open airway) may become necessary
- No vaccine is available
- Croup may be also caused by Respiratory syncytial virus and influenza virus
(account for 1% to 10% of case)

Coxsackie A virus

Cause Herpangina:- a severe and febrile pharyngitis

- associated with fever, sore throat, pain on swallowing, anorexia, and vomiting
- the classic finding is vesicular ulcerated lesions around the soft palate and uvula



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TABLE 75–1 Common Infections of the Upper Respiratory Tract

Infection	Important Pathogens	Treatment
Otitis media	<i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Moraxella catarrhalis</i>	Amoxicillin
Acute sinusitis	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>M. catarrhalis</i>	Amoxicillin if symptoms persist for >10 days
Pharyngitis	<i>Streptococcus pyogenes</i> (group A <i>Streptococcus</i>), viruses (e.g., adenovirus)	Penicillin or amoxicillin if group A <i>Streptococcus</i> diagnosed
Common cold	Rhinovirus, coronavirus, and others	Supportive; zinc may be helpful in reducing duration of symptoms
Croup	Parainfluenza virus	Supportive; corticosteroids and epinephrine if moderate or severe symptoms
Laryngitis	Parainfluenza virus and rhinovirus	Supportive
Epiglottitis	<i>H. influenzae</i> type B	Ceftriaxone